Ultrasound and optoacoustic imaging
Part I: Ultrasound in biology and medicine

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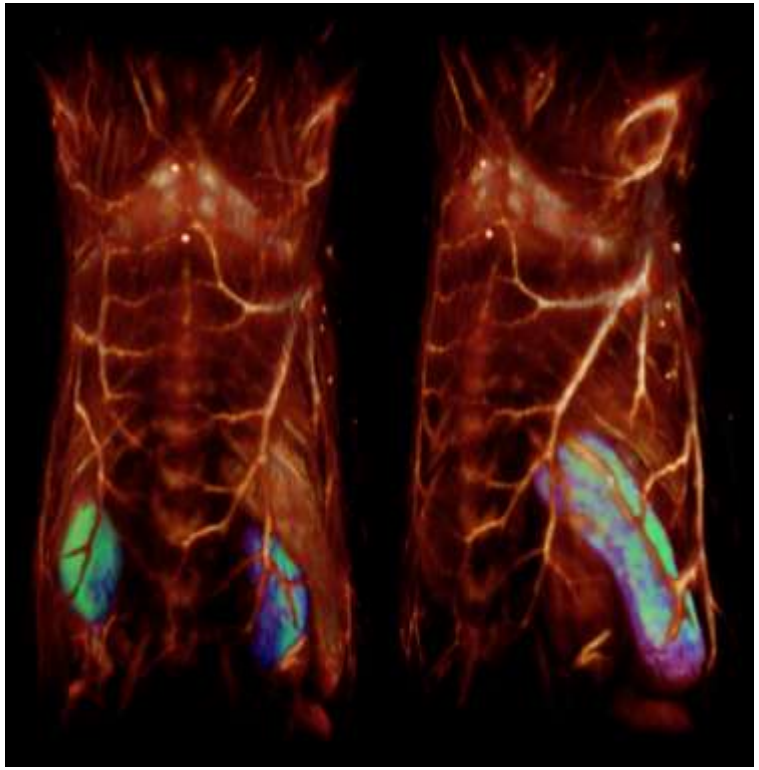
Today's topic



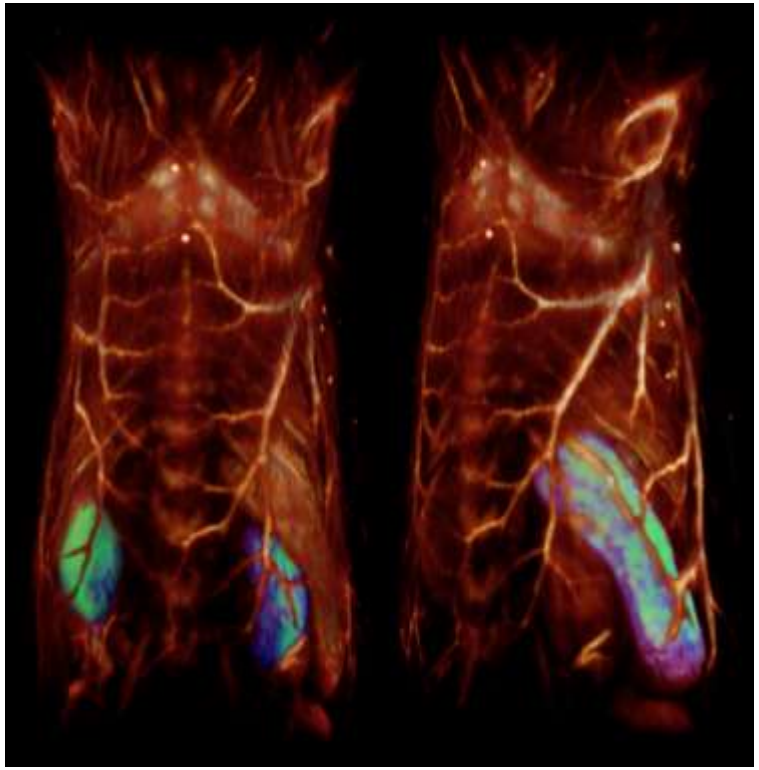
Learning objectives

Part I: Ultrasound in biology and medicine

Part II: Advanced ultrasound imaging

Part III: Optoacoustic imaging

Today’s topic



Learning objectives

Part I: Ultrasound in biology and medicine

Part II: Advanced ultrasound imaging

Part III: Optoacoustic imaging

Ultrasound in biology and medicine

Why do you know about ultrasound in biology and medicine and why do you think it is important?

Let’s go to Wikipedia

Sound is a vibration that propagates as an acoustic wave through a gas, liquid or solid medium

Ultrasound is sound with frequencies greater than the upper audible limit of human hearing in healthy young adults

Acoustic spectrum

The physical principles of acoustic (mechanical) waves apply to any frequency range

Sound/ultrasound in nature

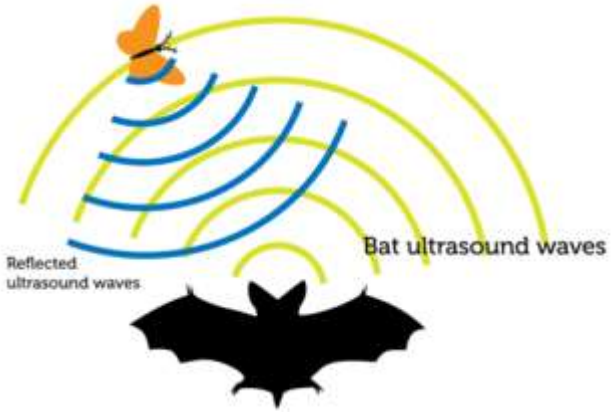
Communication

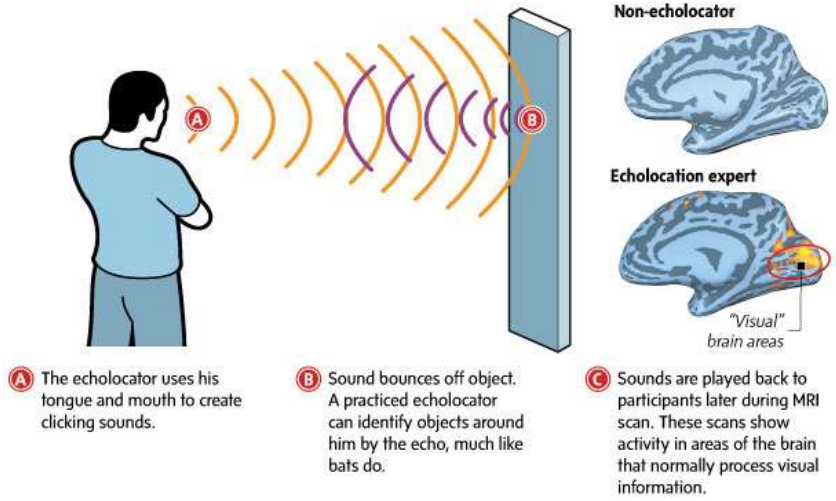
Animals need to send/receive signals

Evolution “expanded” the audible range

Philippine tarsier (Carlito syrichta)

Echolocation

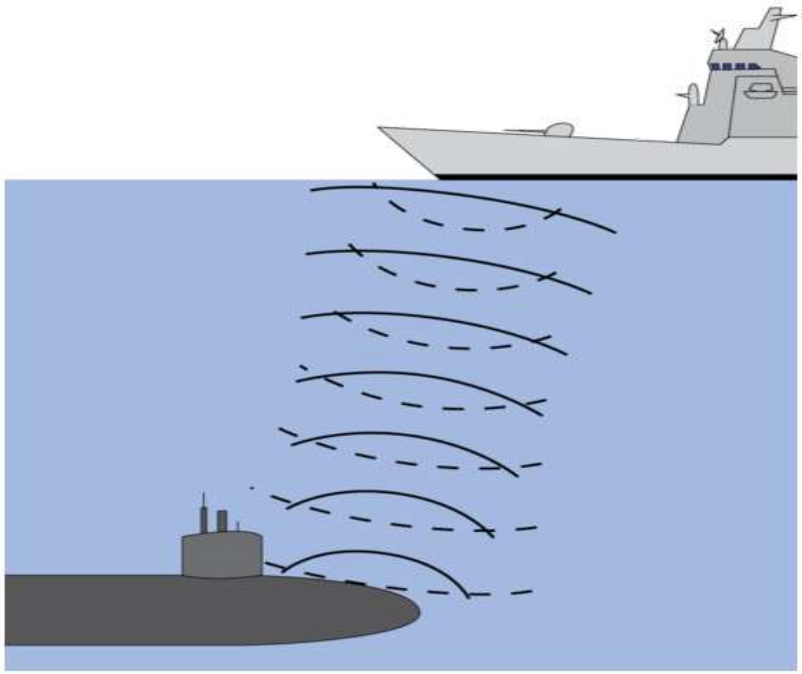




Learning from nature

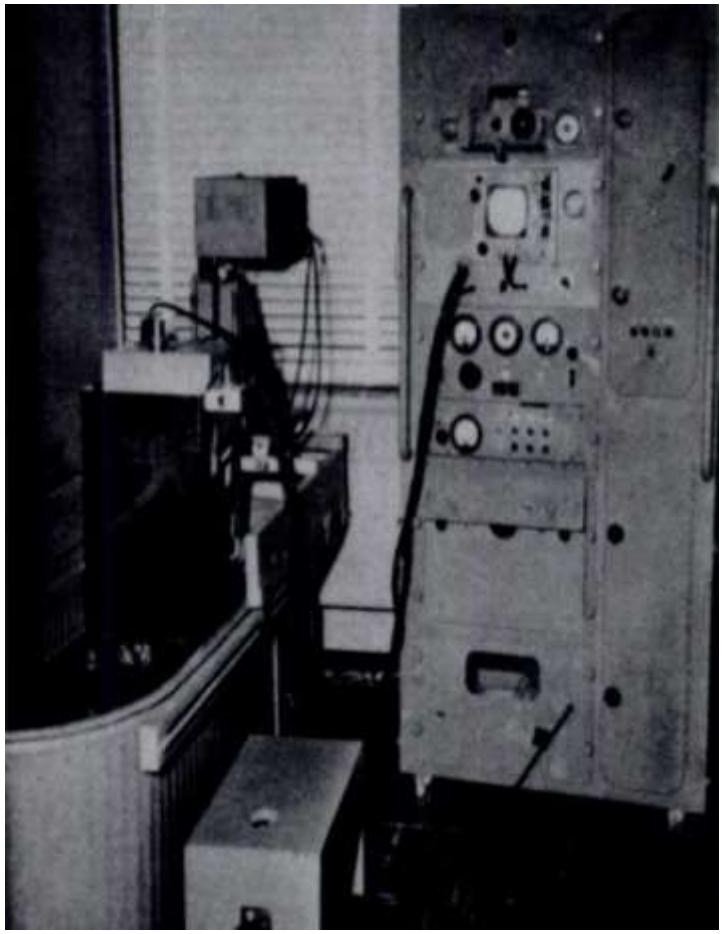
The echolocation principle was implemented with sensors capable of generating and detecting ultrasound

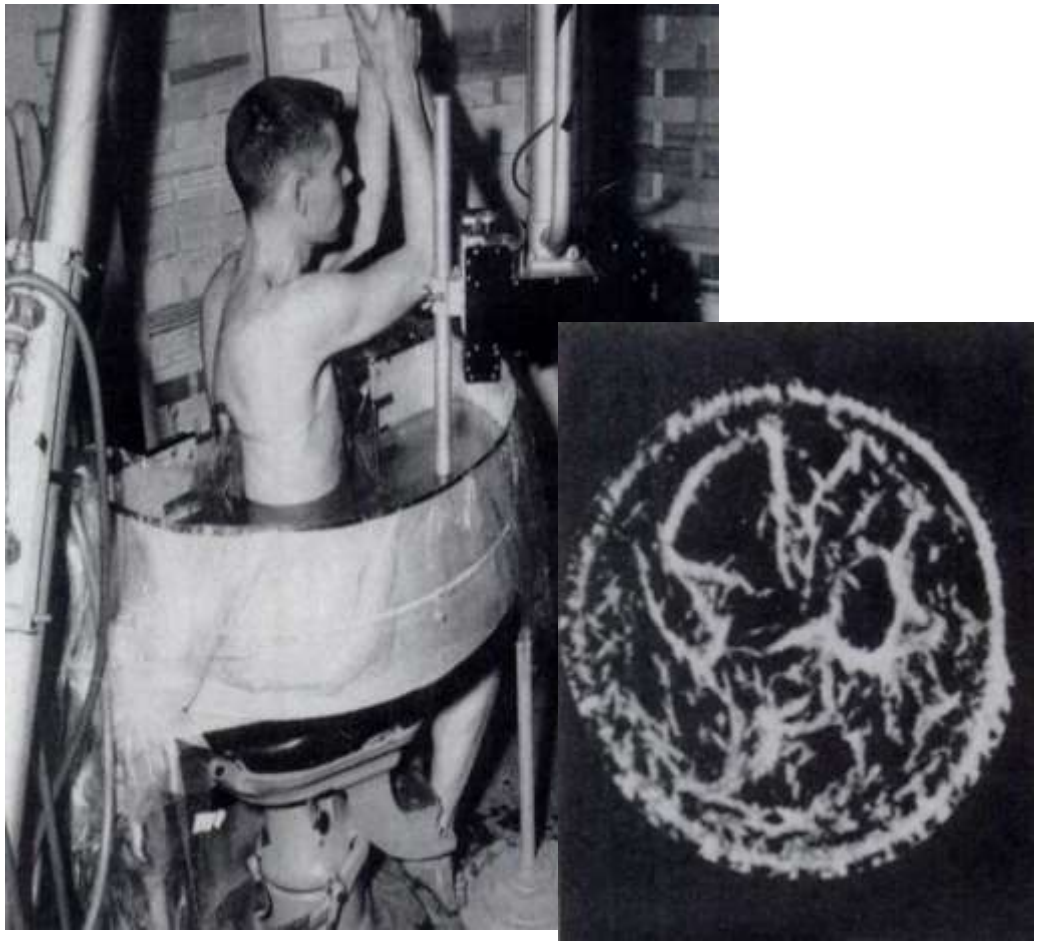
Sonar



Medical imaging

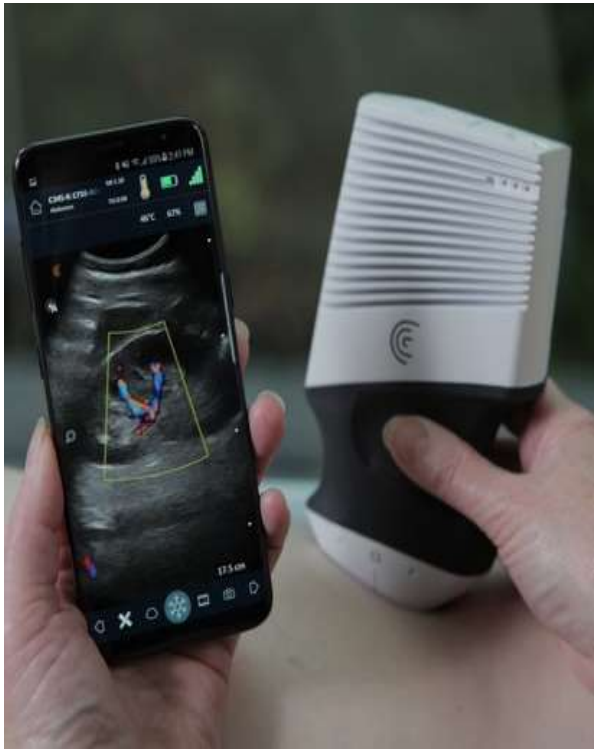
How was it in the early days?

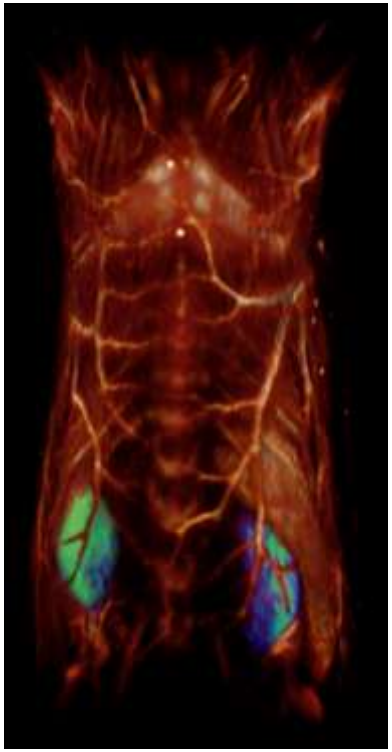




Goldberg et al. AJR Am J Roentgenol 160(1): 189-194 (1993)

How is it today?





Why ultrasound imaging?

Pros

- ✓ Safe (non-ionizing radiation)
- ✓ Real-time operation
- ✓ Portable (bedside/point-of-care testing)
- ✓ Relatively low cost

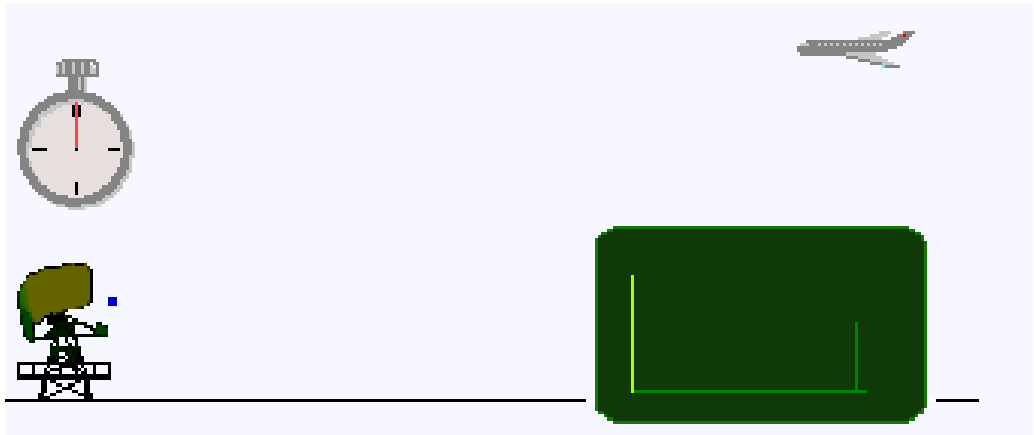
Cons

- Relatively low signal-to-noise ratio (SNR)
- Relatively low penetration depth
- Results can be operator-dependent

Pulse-echo principle

The transmitter (T) generates a pulsed wave that is reflected by an inhomogeneity and obtained by the receiver (R)

Example: Radar (radio detection and ranging)



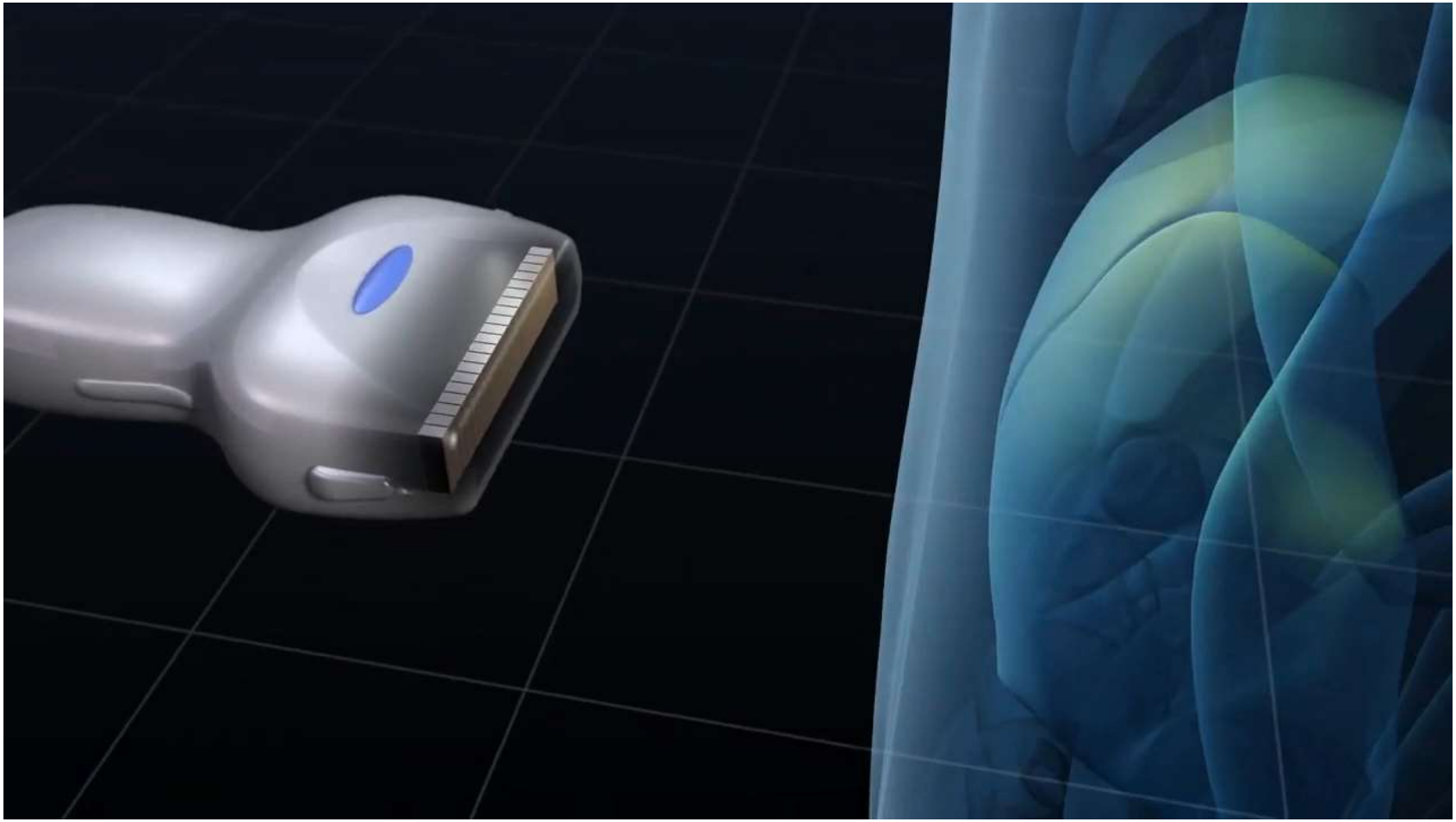
$$\boxed{\mathrm{ToF} = \frac{2d}{c}}$$

ToF – time-of-flight

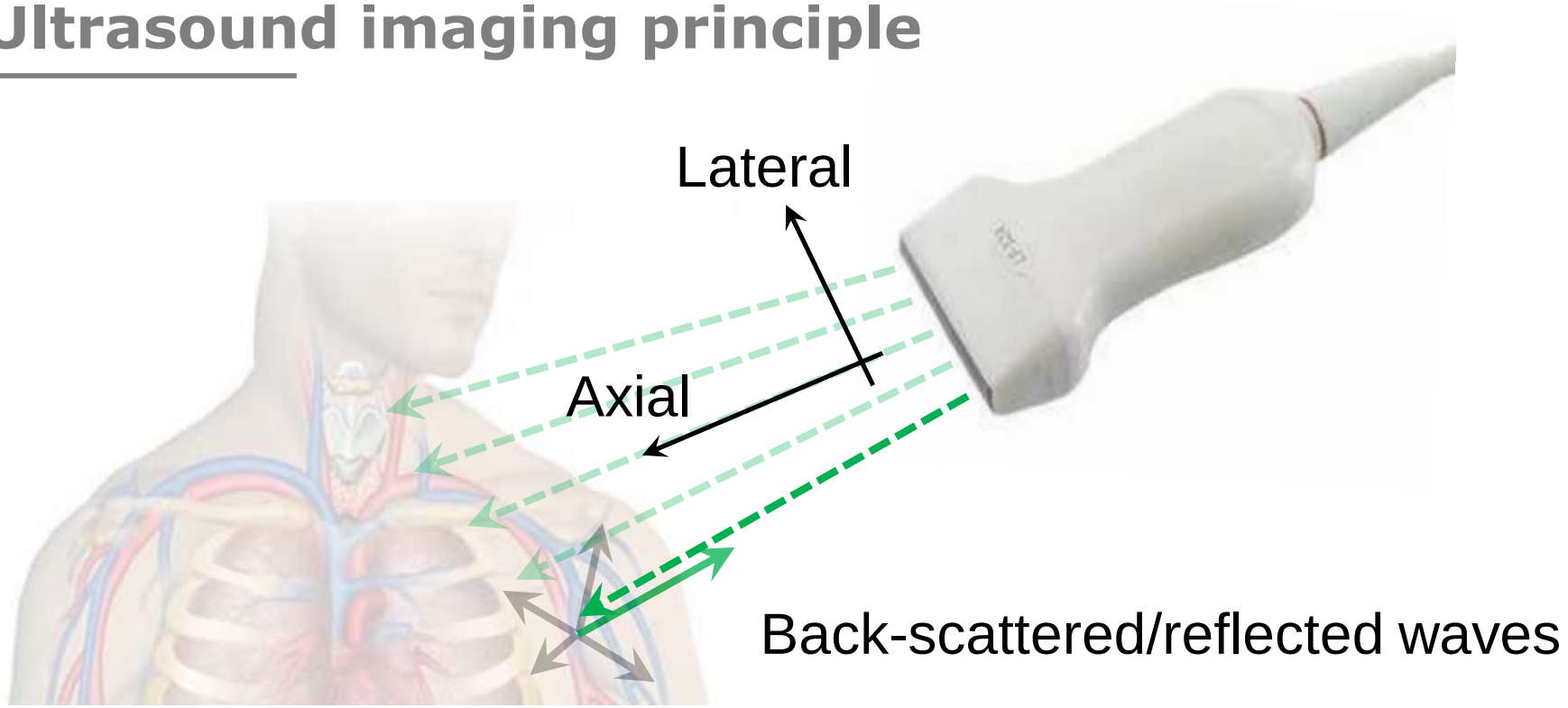
d – distance

c – wave speed

Pulse-echo principle



Ultrasound imaging principle



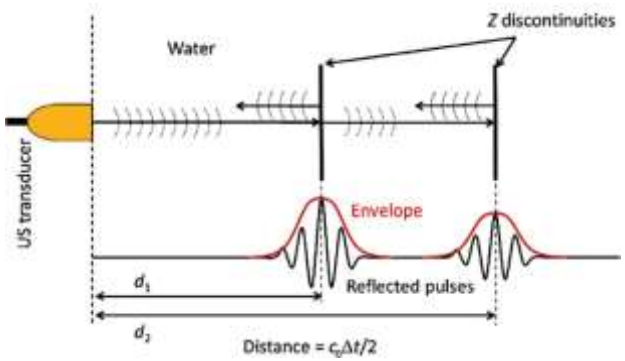
- Resolution in the axial/depth direction is provided by the time-of-flight of ultrasound waves

- Resolution in the lateral direction is provided by the directivity of the ultrasound beam

Imaging modes

Ultrasound imaging modes refer to the different ways information is presented in ultrasound imaging systems

A-mode ('A' for amplitude)

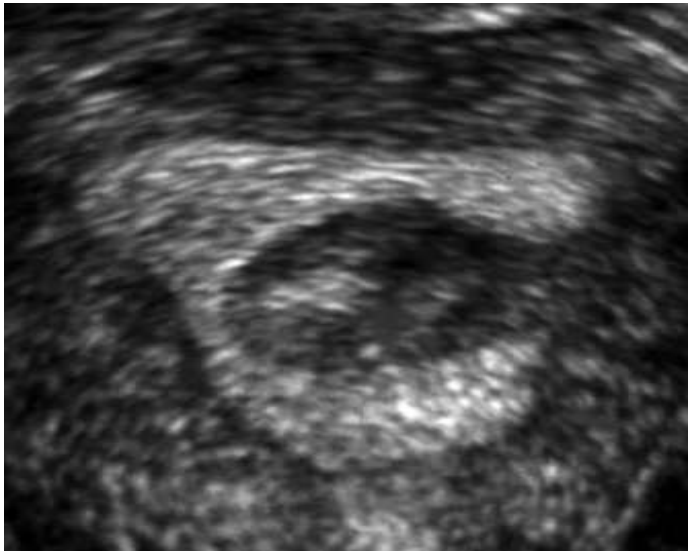


$$d = \frac{ct}{2}$$

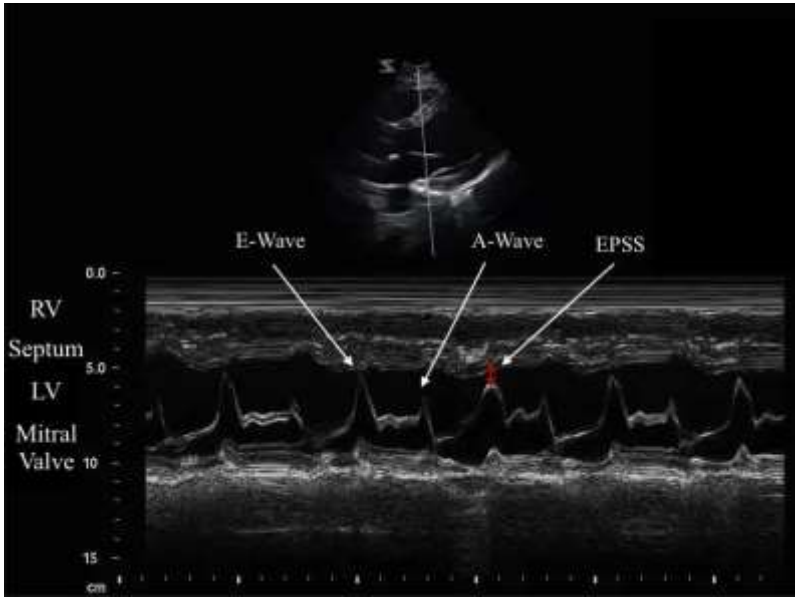
B-mode ('B' for brightness)

Imaging modes

C-mode ('C' for contrast)

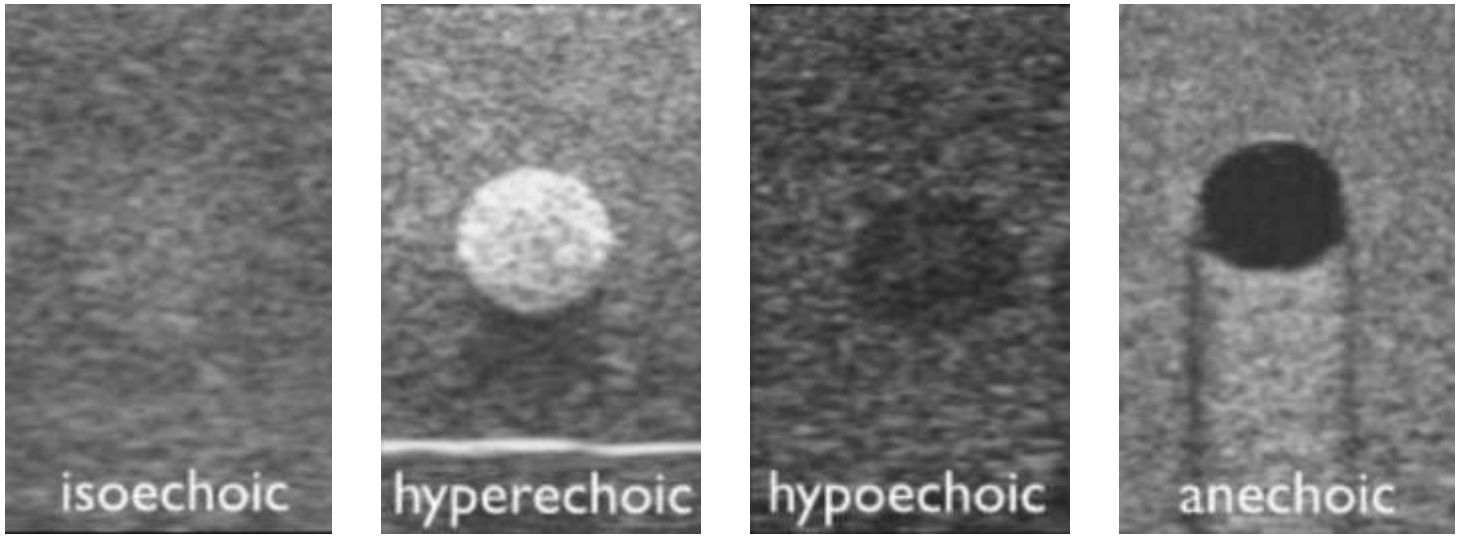


M-mode ('M' for motion)



Echogenicity

Echogenicity refers to the ability of a structure to bounce an echo, i.e., return a signal, and corresponds to image brightness



Isoechoic – same echogenicity as surrounding structures

Hyperechoic – more echogenic than surrounding tissues

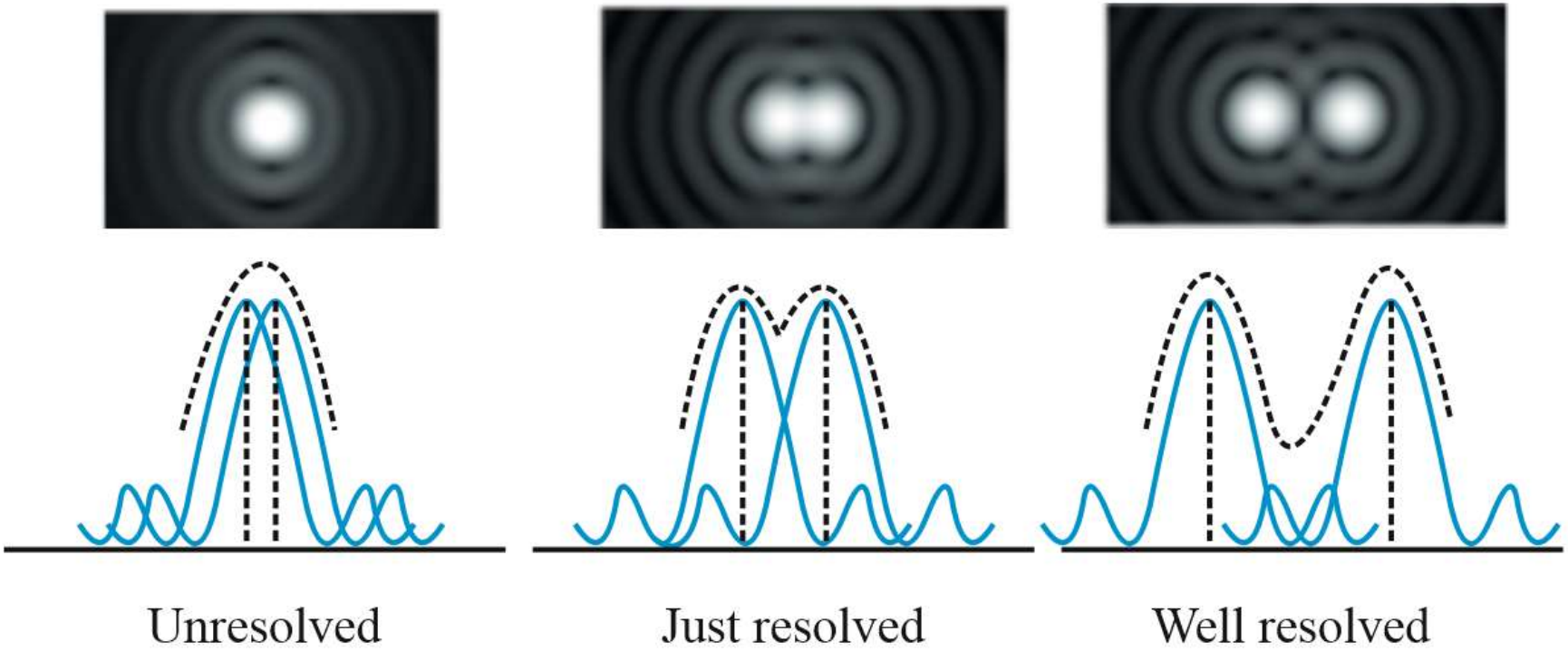
Hypoechoic – less echogenic than surrounding tissues

Anechoic – no internal echoes

What is the resolution?

Spatial resolution

The **spatial resolution** of an imaging system is typically defined as the minimum distance for which two different features can be recorded as distinct within an image



Spatial resolution

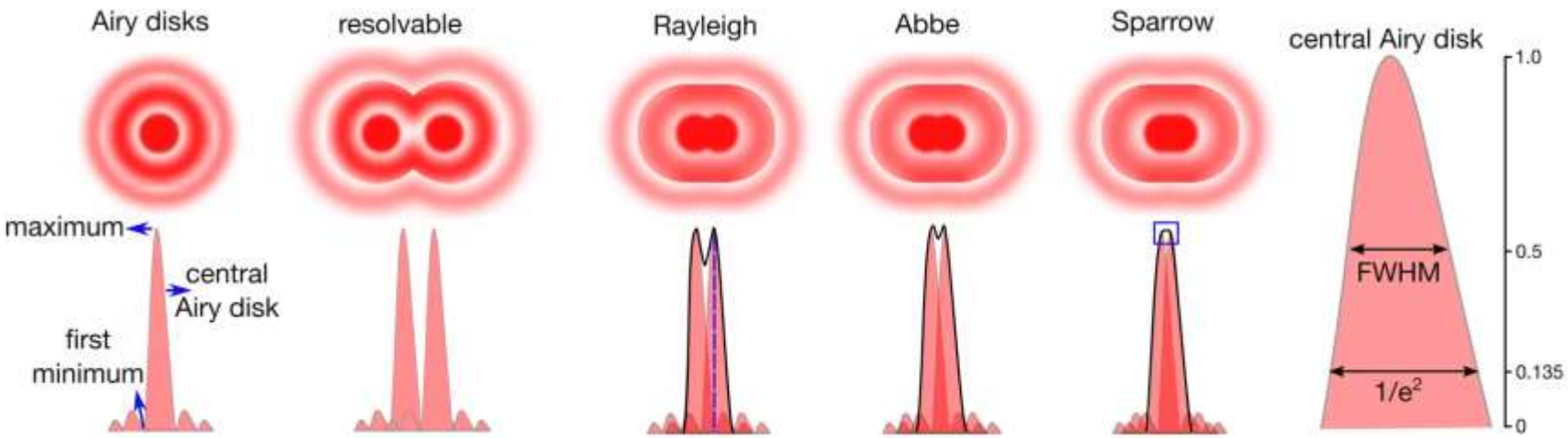
Different criteria are used to quantify the spatial resolution

George Biddell Airy

Lord Rayleigh

Ernst Abbe

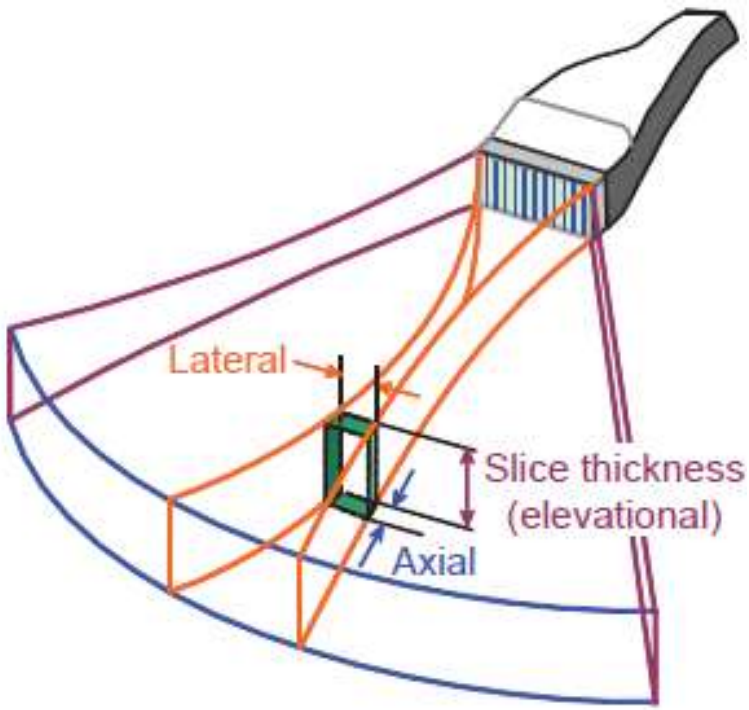
Carroll Mason Sparrow



Minimum distance for which the intensity at the midpoint between two peaks has a minimum

Width of the point spread function (PSF) of the imaging system

Axial, lateral and elevational resolution



Axial resolution

parallel to the ultrasound beam

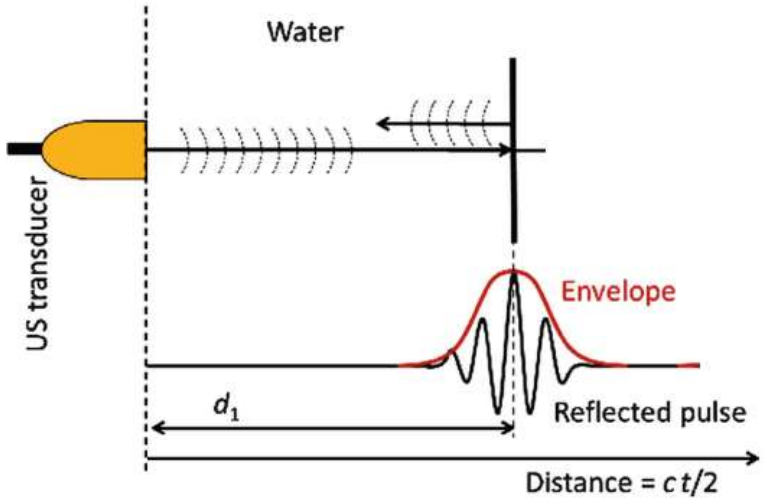
Lateral resolution

perpendicular to the ultrasound beam in the imaging plane

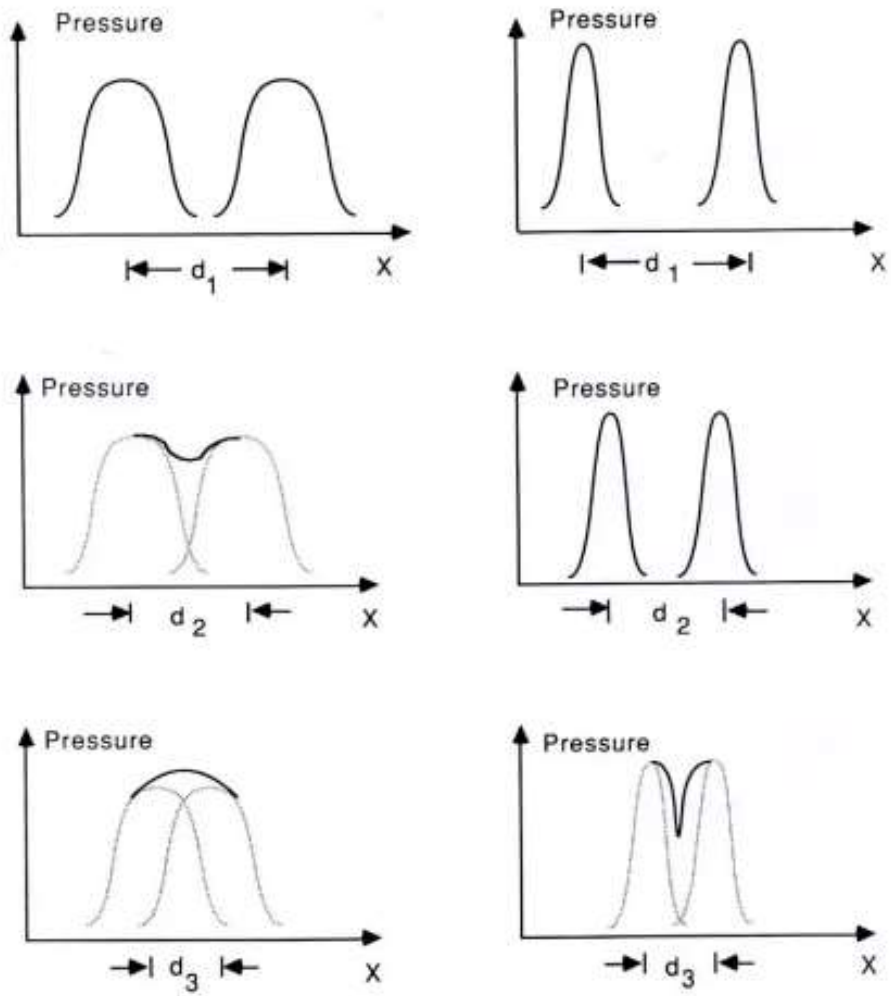
Elevational resolution

perpendicular to the ultrasound beam and the imaging plane, determines the slice thickness

Axial resolution



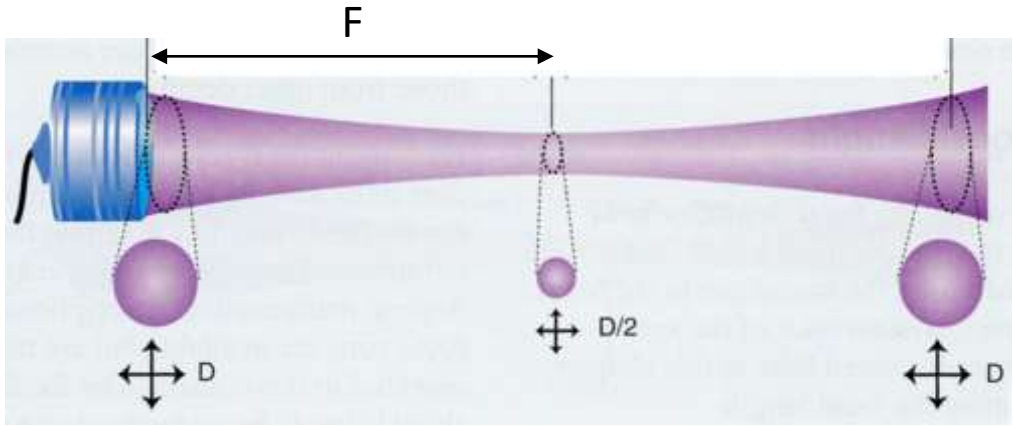
Low frequency (bandwidth) | **High frequency (bandwidth)**



The **axial resolution** can be estimated as the width of the envelope of the signal for a point source

$$r_a = \frac{c\Delta t}{2}$$

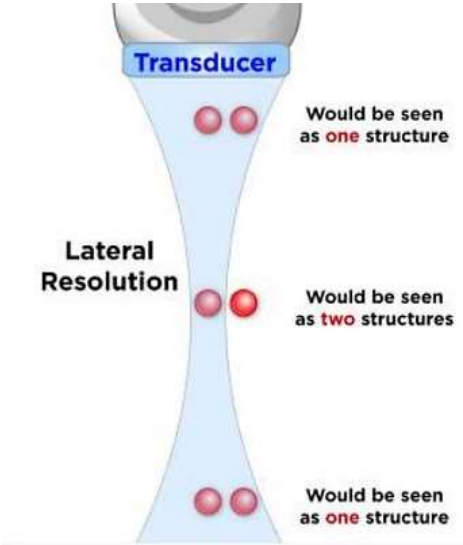
Lateral resolution

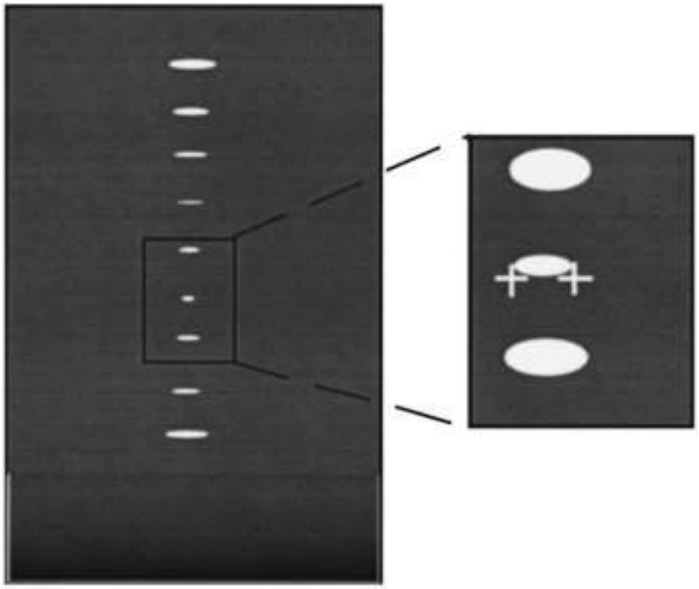


The **lateral resolution** can be estimated as the width of the ultrasound beam

$$r_l = \lambda \frac{F}{D} = \frac{\lambda}{2\,\mathrm{NA}}$$

NA – numerical aperture





Contrast resolution

Contrast resolution (or simply **contrast**) refers to the ability of an ultrasound system to differentiate between tissues having different characteristics

Contrast-to-noise ratio

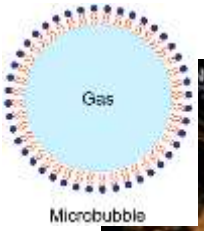
$$\mathrm{CNR} = \frac{|S_\mathrm{A} - S_\mathrm{bg}|}{\sigma_\mathrm{bg}}$$

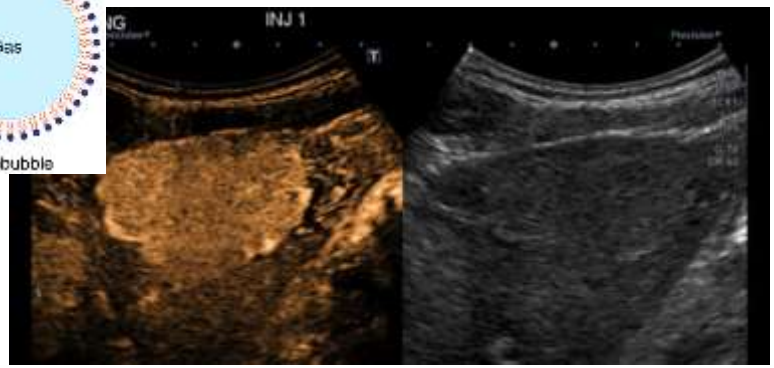
S_A - signal region A

S_bg - signal background

σ_bg - standard deviation background

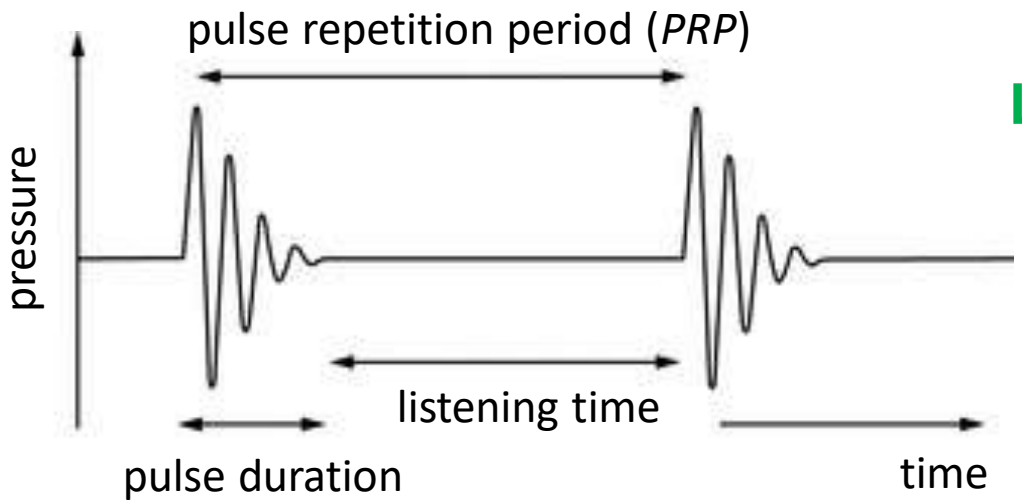
Contrast can be enhanced by injecting a contrast agents. Microbubbles are clinically approved for this purpose





Temporal resolution

The **temporal resolution** is defined as the amount of time needed to revisit and acquire data for the exact same location



Listening time

must cover echoes from the deepest structures (depth *d*)

$$PRP >= \frac{2d}{c}$$

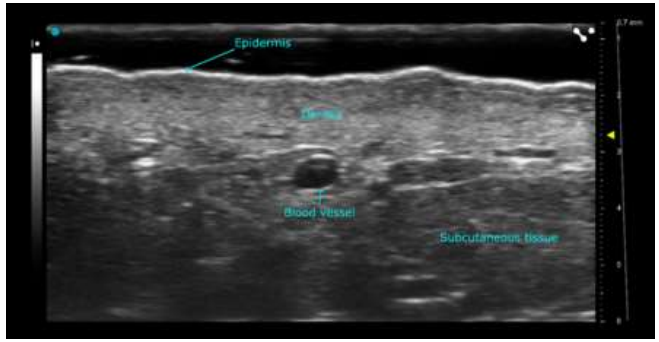
The achievable temporal resolution depends on the number of A-mode signals (*N*) required to form an image

$$r_t = N\frac{2d}{c}$$

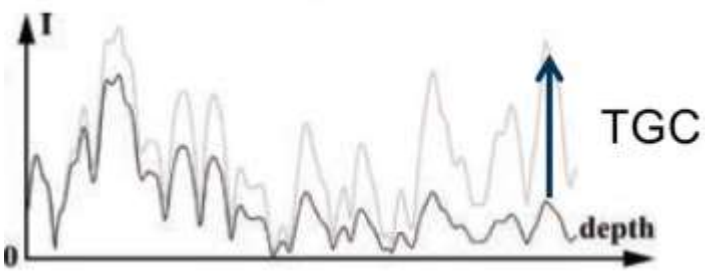
Penetration depth

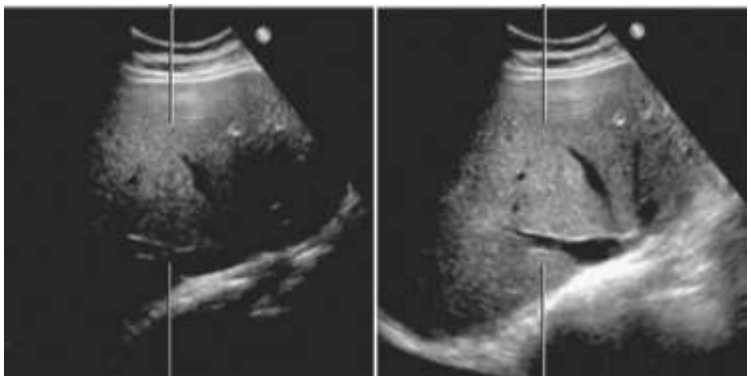
The **penetration depth** in ultrasound imaging is limited by frequency-dependent acoustic attenuation to approximately 200 wavelengths

$$\alpha(f) = \alpha_0 f^n$$



Time-gain-compensation (TGC) is applied to account for tissue attenuation by increasing the signal intensity with depth

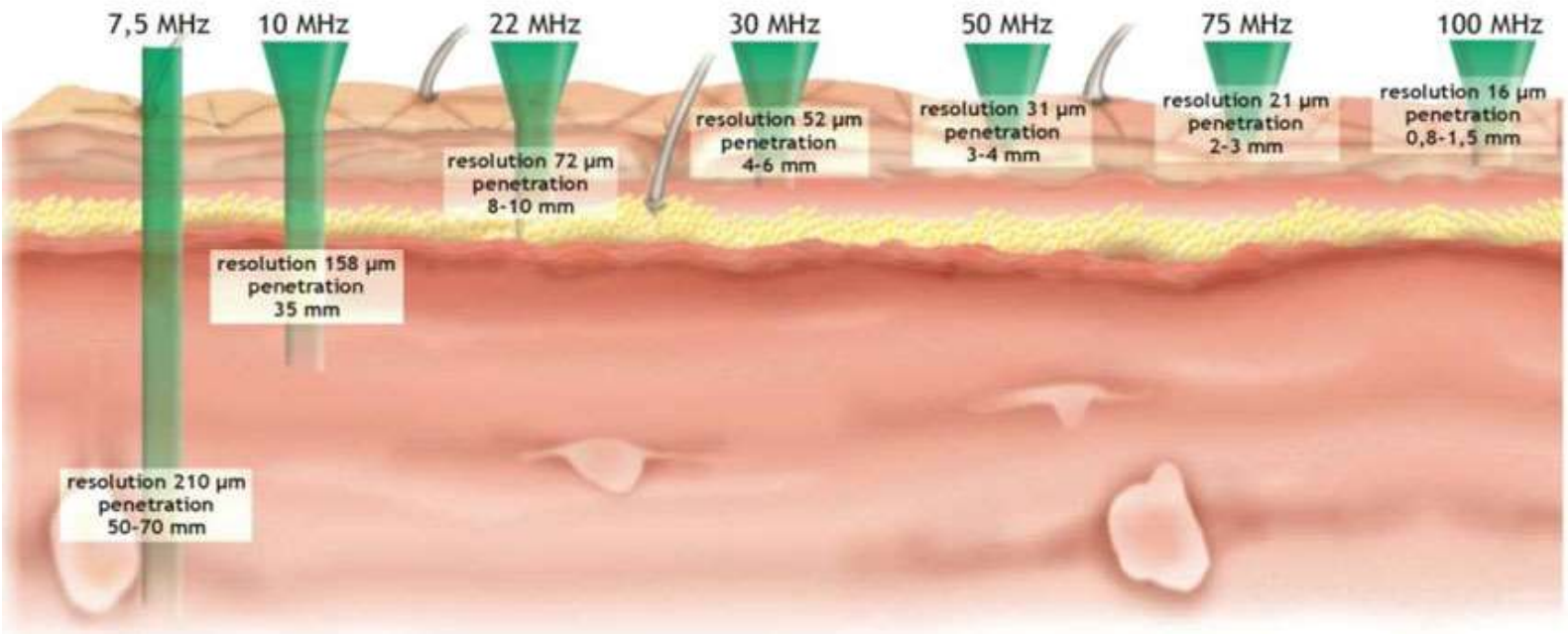




The resolution-depth tradeoff

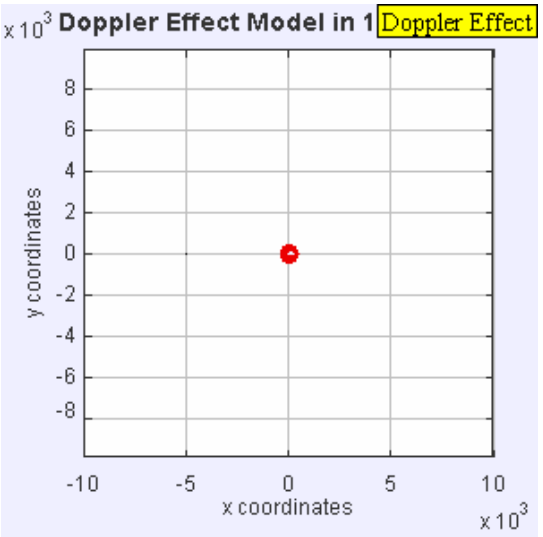
Depth Vs Resolution

Low frequency	High frequency
Low resolution	High resolution
Longer depth	Shorter depth



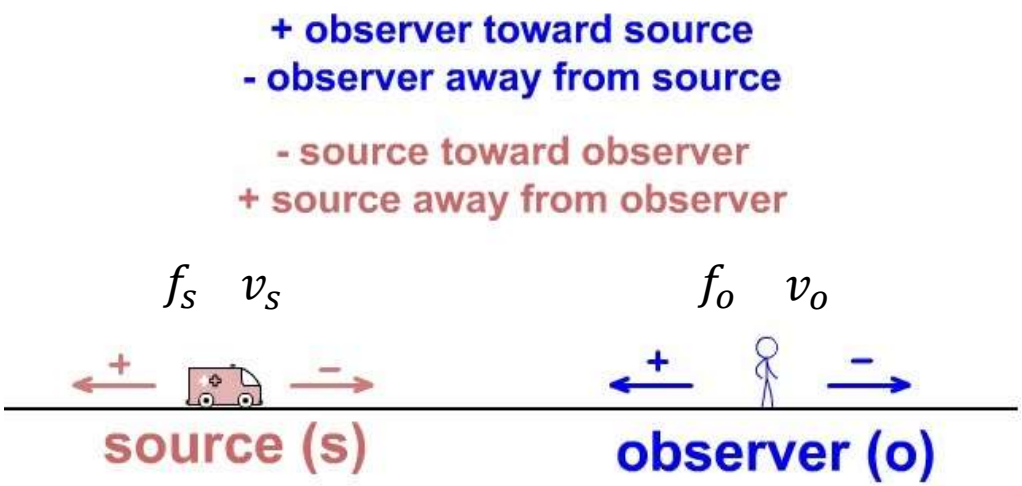
The Doppler effect

The **Doppler effect** is the apparent change in frequency of a wave in relation to an observer moving relative to the wave source



Christian Doppler

The observed effect depends on the radial velocities of the source and the observer

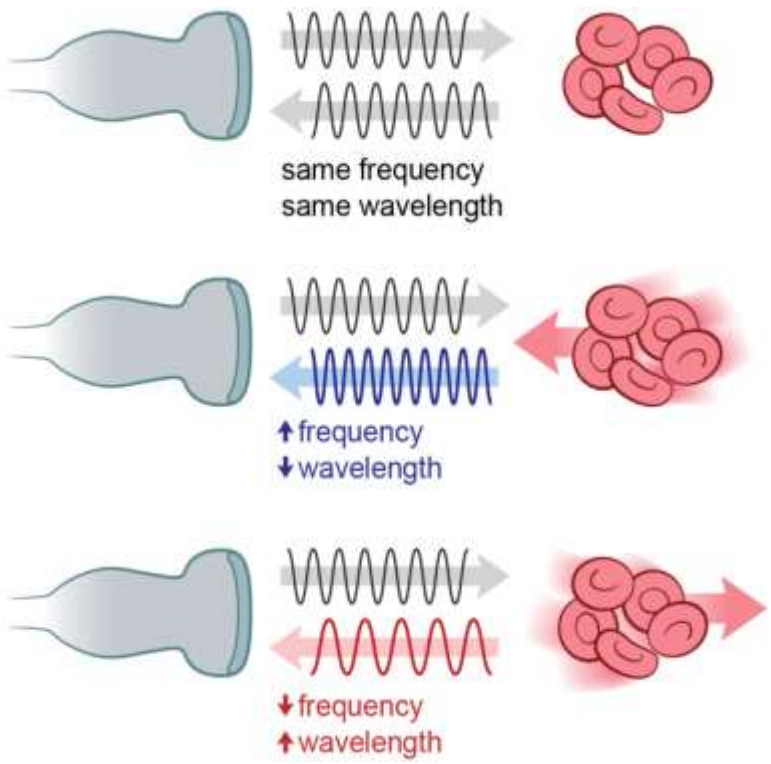


Observed frequency

$$f_o = f_s \left[\frac{c \pm v_o}{c \mp v_s} \right]$$

Continuous wave (CW) Doppler

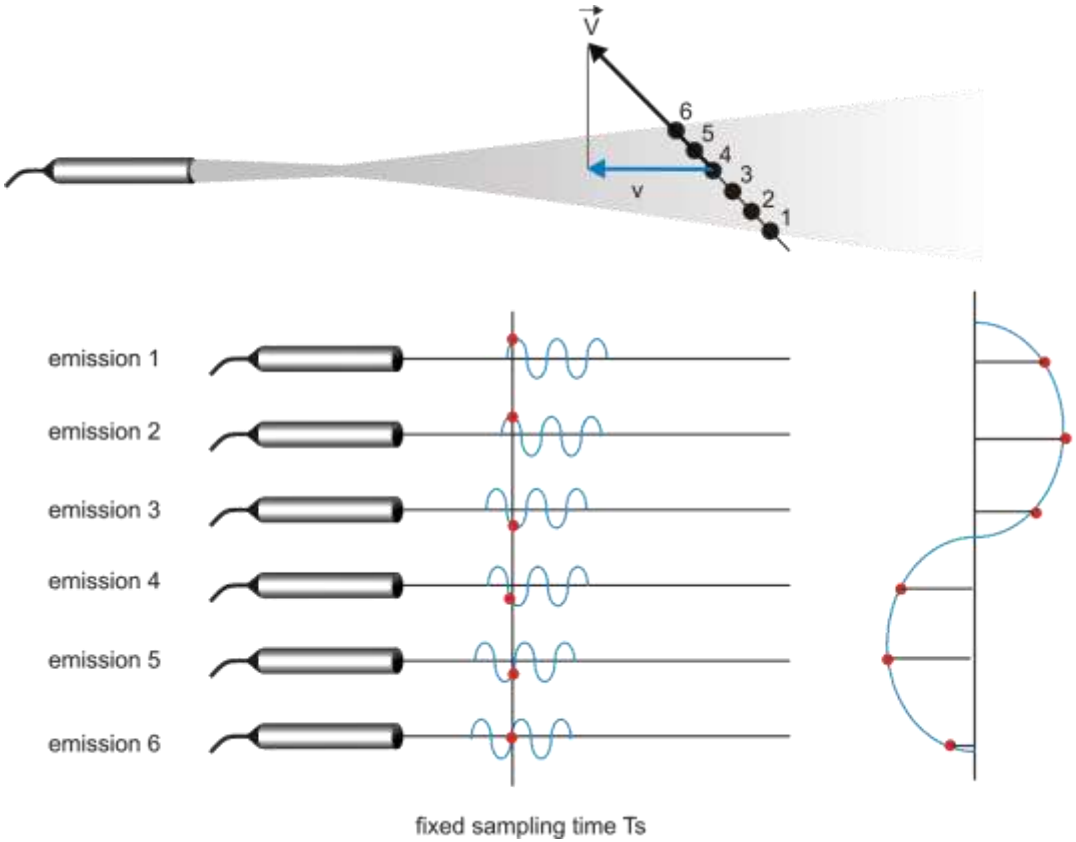
In **continuous wave (CW) Doppler**, two piezoelectric elements are used to emit a time-harmonic wave and receive the echoes, respectively



Doppler-shifted frequencies are in the audible range for typical blood flow velocities and ultrasound transducers around 1-15 MHz

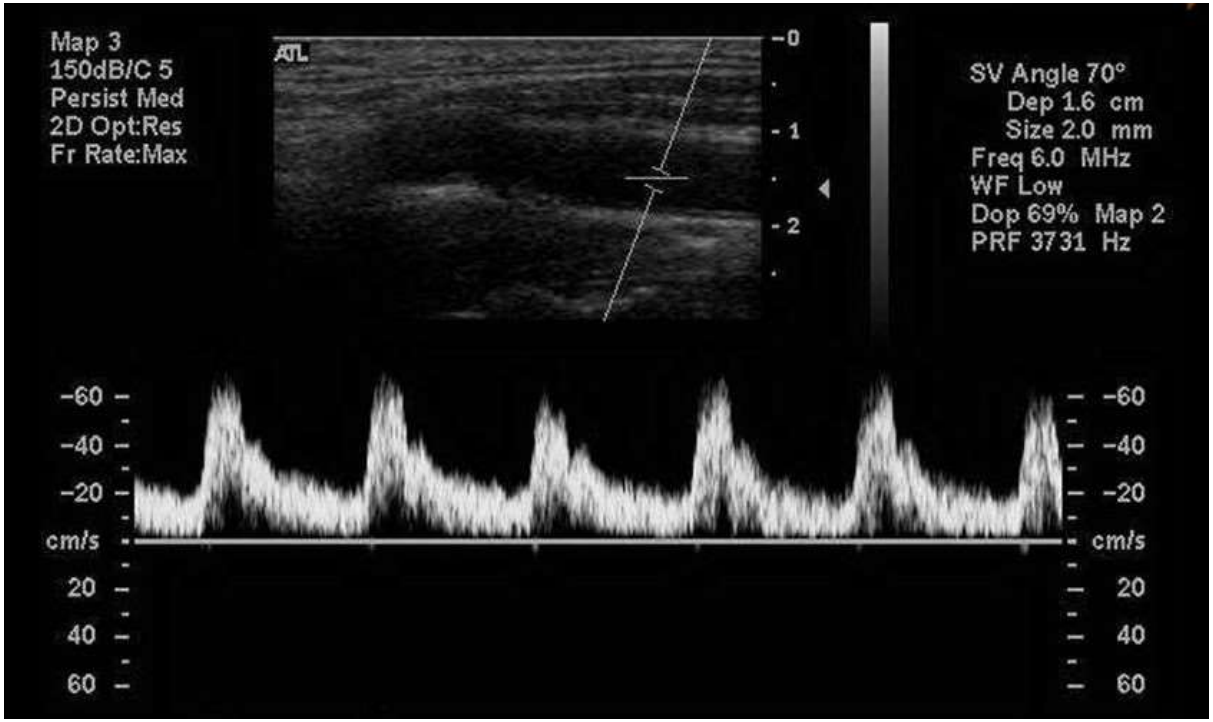
Pulsed wave (PW) Doppler

In **pulsed wave (PW) Doppler**, the frequency of the repeatedly sampled signal at a given point correspond to the velocity at a specific depth



Duplex scanner

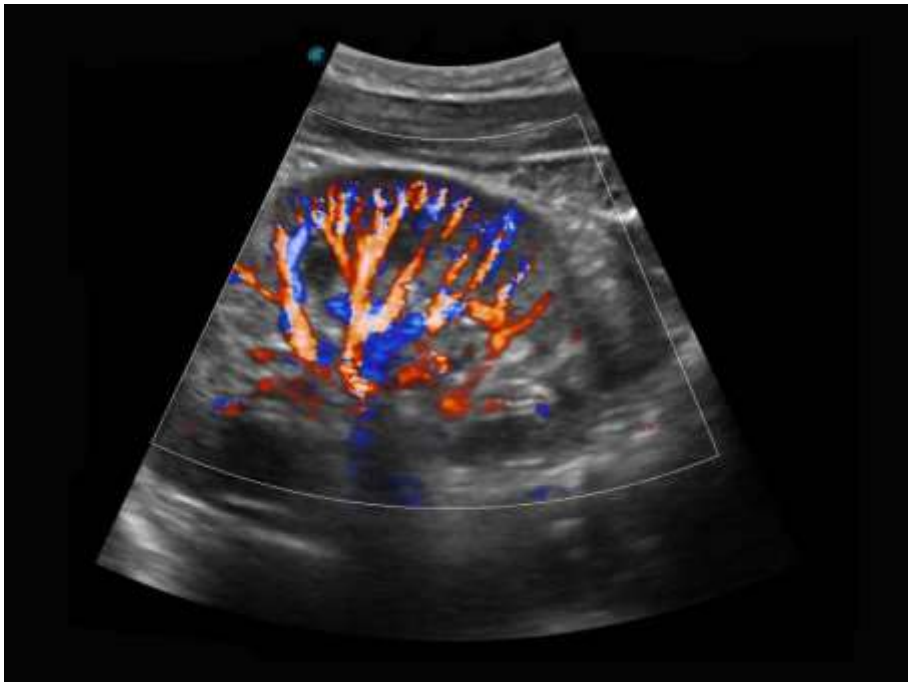
In a **duplex scanner**, a Fourier transform algorithm computes the Doppler spectrum for a ~5-10ms time window and is displayed in real time



The angle of the vessel relative to the interrogation beam can be estimated in the B-mode image for quantitative velocity calculations

Color Doppler

Color Doppler is a duplex imaging mode in which the speed and direction of blood flow are shown in different colors in real time



The BART mnemonic

Blue away, red towards

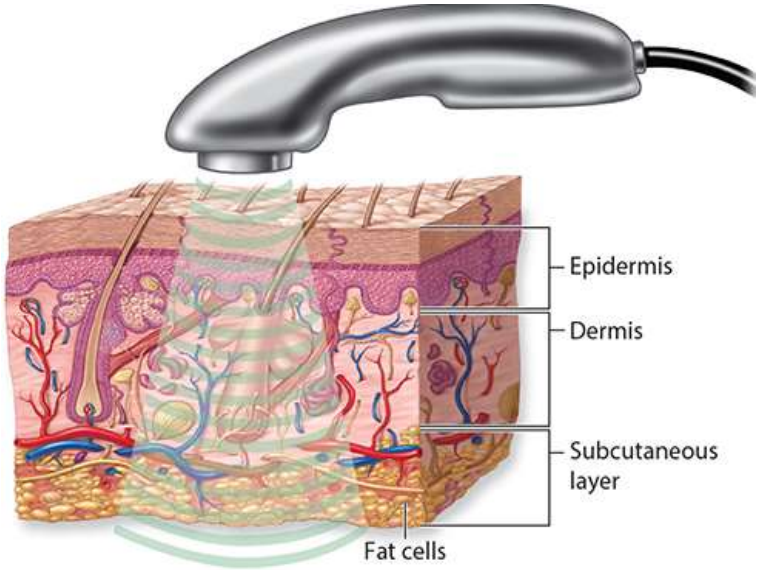
Power Doppler

In **power Doppler**, the colormap displays the integrated power in the Doppler signal rather than the mean frequency shift

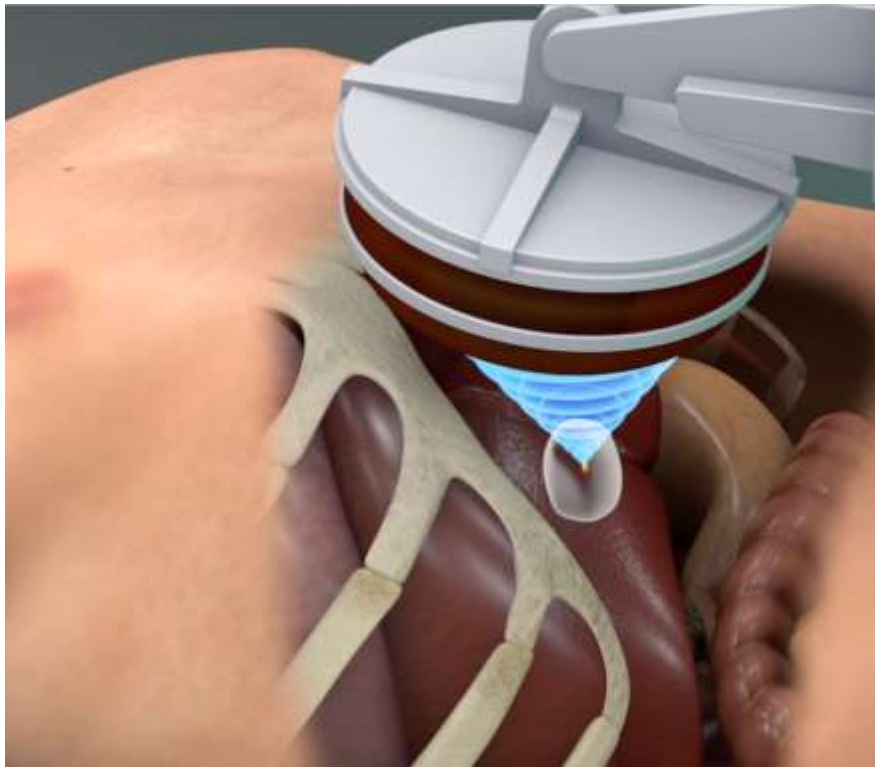
The sensitivity to low flow within a tissue is enhanced

Ultrasound therapy

Low-intensity pulsed ultrasound (LIPUS)

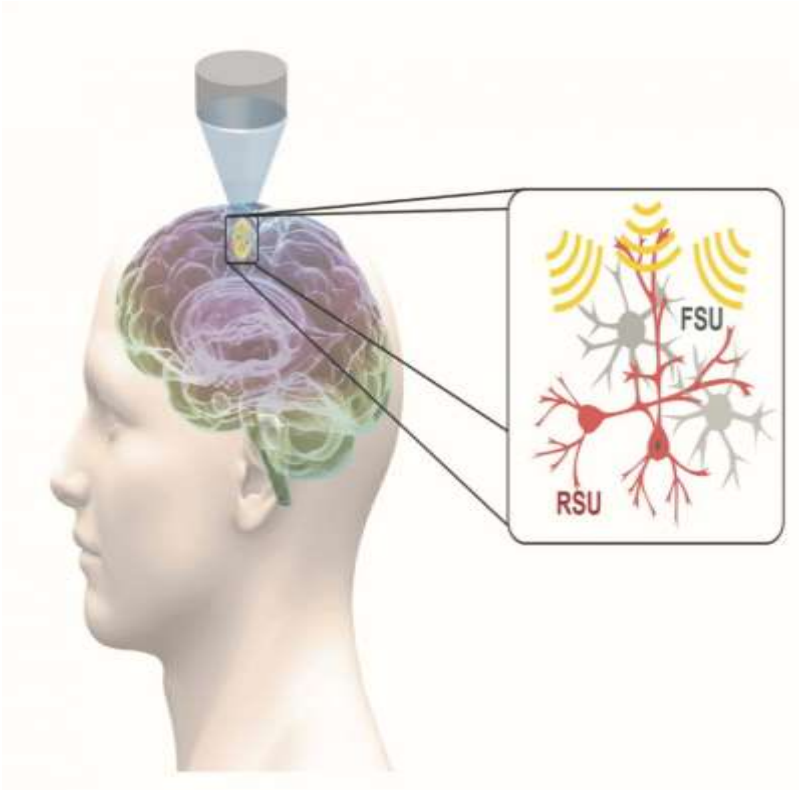


High-intensity focused ultrasound (HIFU)

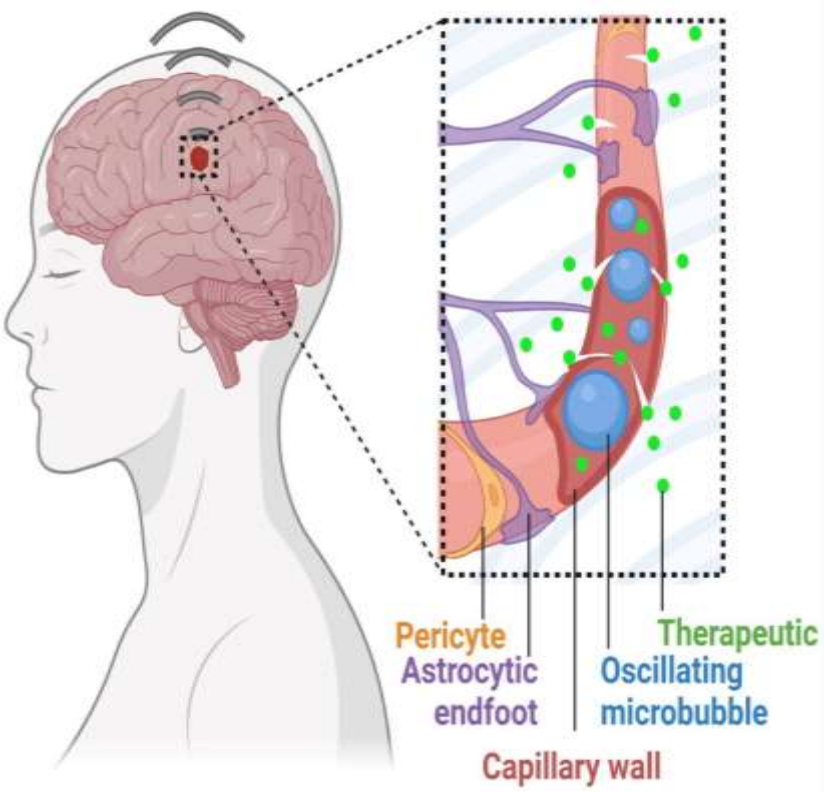


Ultrasound therapy

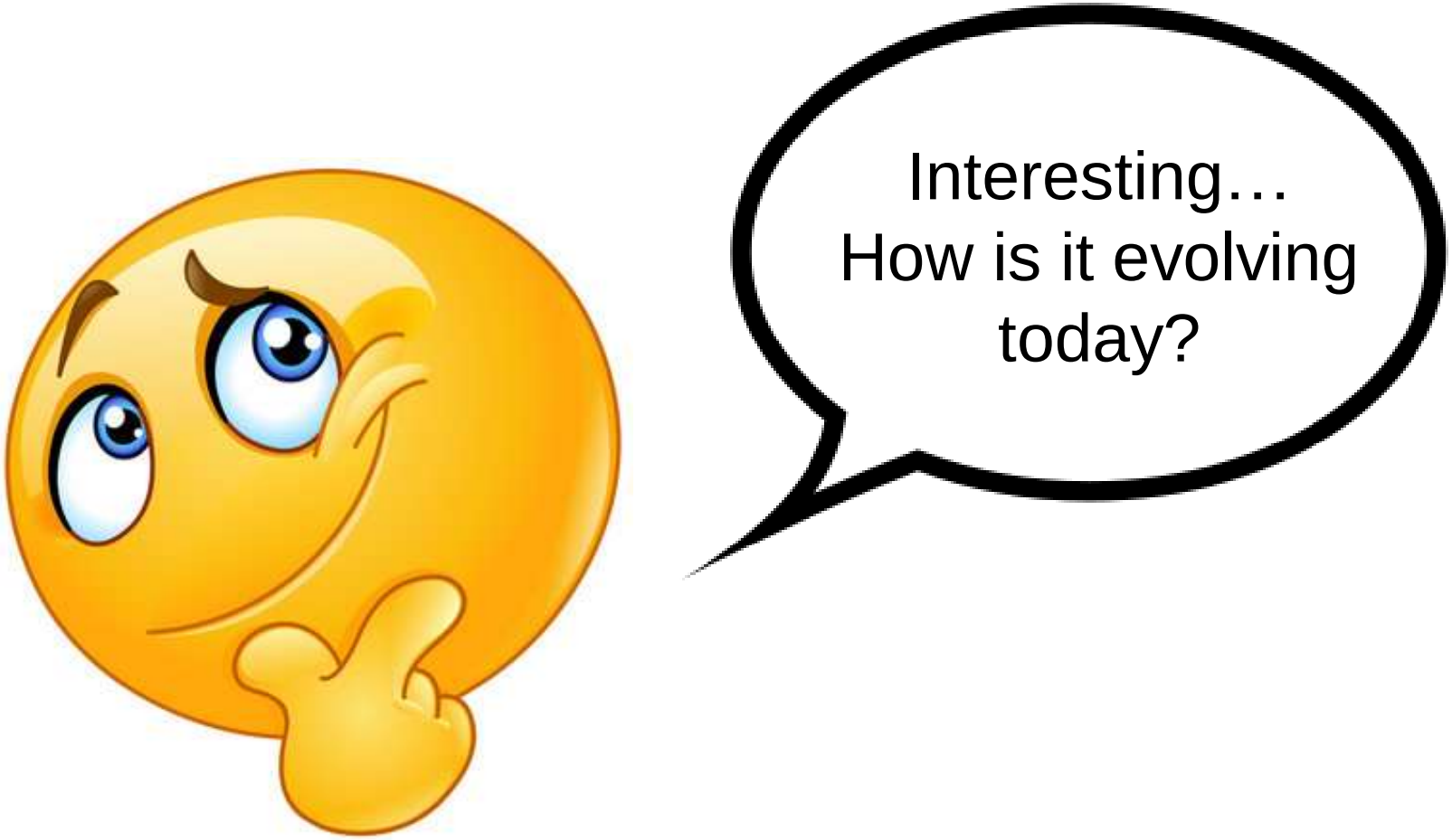
Ultrasound Neuromodulation



Blood brain barrier (BBB) opening



What is next?



Thank you